Murray State University

Types of AI

Demo 1 - rules, machine learning, neural networks - 20 minutes

You need: one browser and the demo's `index.html`. It works offline. Start with a fresh page; use Settings -> Presentation mode to project, Open Presenter Notes for this guide in the app, and Open printable PDF when a handout helps.

The question for all 20 minutes

Where does this opponent's skill come from? The board and game rules stay the same. What changes is the source of the move: written instructions, experience stored as move scores, or a network's adjustable numbers-called weights-reused to score many board positions.

> *"Do not rank these as old AI and new AI. Watch what supplied the skill."*

1a - Rules-based intelligence - 5 minutes

Question

Can an opponent be skilled when it has never learned from a game?

Activity

1. Open **1a Rules-based intelligence** and play a few moves against All 8.
2. Point to the highlighted rule after each reply. Switch to First 2 and invite the room to find a fork, then return to All 8.
3. Choose the policy check if time allows. It examines every reachable game line for this written policy.

Expected observation

The opponent can explain a move by naming a rule. All 8 has no losing line because the app checks the whole policy, not because it won a few demonstrations.

Teaching limit

Its skill came from a person's move priorities. Changing its behavior means revising those instructions; playing it supplies no new experience.

1b - Machine learning - 7 minutes

Question

What changes when an opponent keeps a score for positions it has experienced?

Activity

1. Open **1b Machine learning**. In Era 0, use **Show move scores**; every score begins at zero.
2. Play one short game, then run a visible training burst. Select an earlier and a later era to compare their move scores.
3. Use the policy check on a later era if time allows, then use **New game** to return the board to play.

Expected observation

Completed games change stored move scores. A later era can choose differently because the table remembers estimates for positions it has encountered.

Teaching limit

Human play does not train this table. Only the app's training burst updates it. The table also has one separate score per position, so it does not automatically carry what it learned to a different situation.

1c - Neural networks - 7 minutes

Question

Can adjustable numbers reused across many boards approximate the scores a table learned?

Activity

1. Open **1c Neural networks** and choose **Create learning examples**. A fresh table learner plays 20,000 seeded games; its resulting estimates are frozen as the teacher.
2. Choose **Train network**. Compare the error for examples used in training with the error for positions kept out of training.
3. Read the separate playing score, then use **Show move scores** to see the network's calculation for each legal move.

Expected observation

Training changes weights: adjustable numbers reused to calculate scores for many boards, rather than a separate stored entry for every board. Lower training and held-out error mean closer agreement with this frozen teacher; the playing score is a different measurement.

Teaching limit

Preparation creates a fresh learned teacher; it does not copy answers from a perfect-game solver. People still chose the board description and learning procedure. Held-out error measures agreement on withheld examples, not playing strength or a promise about every unseen board. When the network plays, it uses its own calculation and does not consult the frozen table.

A fresh teacher can leave an unvisited position at its initial score of zero. That is a limit of the examples, not proof that the position is neutral in perfect tic-tac-toe.

> *"A neural network is a type of machine learning. Here it learns a compact approximation of another learner's experience."*

Optional advanced extension - 1 minute

Open Ultimate tic-tac-toe from the neural tab only after the three-stage comparison. It adds facts a one-board picture omits: a board's place in the larger grid and where a move sends the next player. Network weights cannot repair missing inputs; a network given the same incomplete one-board picture misses those facts too.

Close with evidence, not a winner

- Which source of skill was easiest to explain? Which was easiest to update?
- Why compare held-out error with training error?
- What evidence would you need before calling a network unbeatable?

Reopen Guide with the header's Guide button. Settings also offers Open Presenter Notes, Presentation mode, and Reset; Reset returns the full demonstration to its initial state.

Bryant Harrison - Murray State University - Types of AI - single-file offline demo - no accounts or runtime network requests